

A Quantitative Analysis of Time-Varying Fiscal Multipliers and Their Determinants *

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Abstract

This paper studies how the effects of public spending on real economic activity in Peru evolved between the early 1990s and 2017. I estimate a time-varying parameter VAR with stochastic volatility and identify structural shocks using sign restrictions that leave the response of GDP to fiscal shocks unrestricted. The analysis proceeds in two steps. First, I estimate the time-varying response of GDP to public spending shocks and construct impact, cumulative, and present-value spending multipliers. Second, I examine how the estimated multipliers covary with macroeconomic and fiscal conditions. The results indicate substantial time variation in fiscal transmission: the short-run impact of public spending rose through the late 1990s and declined thereafter. In the second-stage regressions, higher public debt-to-GDP ratios are associated with smaller estimated spending multipliers.

Keywords: Public spending, fiscal multiplier, sign restrictions, TVP-VAR-SV, Bayesian econometrics.

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1 Introduction

The Peruvian economy experienced large macroeconomic and institutional changes over the sample period considered in this paper. These include the Asian–Russian crisis of the late 1990s, the 1998 El Niño event, the global financial crisis, and major changes in the fiscal framework.¹ Such changes make the assumption of a constant fiscal transmission mechanism particularly restrictive.

Existing work for Peru already suggests that the effects of fiscal policy are not stable across periods or states of the economy. Mendoza and Melgarejo (2008) find stronger fiscal effects during 1990–2006 than during the 1980s, a period they associate with stronger public finances. Sánchez and Galindo (2013) and Salinas and Chuquilín (2013) report larger spending effects in low-growth or recessionary states. Most of this evidence, however, relies on subsample comparisons or a small number of discrete regimes. Fiscal transmission may instead evolve gradually as the macroeconomic environment, institutions, and financial conditions change.

This paper asks two questions. First, how has the response of economic activity to public spending shocks evolved over time in Peru? Second, which macroeconomic and fiscal conditions are associated with the evolution of the spending multiplier?

To answer the first question, I estimate the time-varying parameter VAR with stochastic volatility (TVP-VAR-SV) of Primiceri (2005). The model allows both coefficients and shock variances to evolve over time. Structural shocks are identified with sign restrictions in the spirit of Mountford and Uhlig (2009). For the fiscal shocks of primary interest, only the response of the fiscal variable itself is restricted on impact, leaving the response of GDP to be determined by the data. I then use the time-varying impulse responses to construct impact, cumulative, and present-value spending multipliers.

To address the second question, I relate the estimated multipliers to the business cycle, financial conditions, credit growth, public debt, trade openness, and the savings rate. This second stage is descriptive rather than causal. The multipliers are generated regressands, and lagging the covariates does not by itself eliminate endogeneity or reverse causality. The results should therefore be interpreted as evidence on conditional comovement and as a guide to mechanisms that may merit further causal analysis.

The paper contributes to a relatively small literature that studies fiscal transmission with time-varying parameter models, including Kirchner, Cimadomo, and Hauptmeier (2010), Pereira and Lopes (2010), Berg (2015), and Glocker et al. (2017). It also complements the Peruvian literature by allowing fiscal responses to vary continuously over time rather than imposing fixed subsamples or a small number of regimes.

The remainder of the paper is organized as follows. Section 2 reviews the related literature. Section 3 presents the empirical methodology. Section 4 describes the identification strategy, and Section 5 discusses the data. Section 6 presents the results and sensitivity analysis. Section 7 concludes.

¹The Fiscal Prudence and Transparency Law was introduced in 1999 and replaced by the Fiscal Responsibility and Transparency Law in 2003. A structural fiscal framework was adopted in 2013 and subsequently modified in 2016. See Ganiko and Montoro (2017).

2 Literature Review

The theoretical effect of government spending depends on assumptions about household behavior, nominal rigidities, financing, and access to credit. In standard Keynesian models, higher government spending raises aggregate demand and output, and the induced increase in disposable income can amplify the initial effect. Ricardian or neoclassical mechanisms can attenuate this response because households anticipate future taxation or because higher government demand affects labor supply, real wages, and private expenditure (Barro, 1974; Aiyagari et al., 1992; Baxter and King, 1993). In New Keynesian models, nominal rigidities support a positive output response, while the presence of liquidity-constrained households can substantially strengthen the consumption channel (Linnemann and Schabert, 2003; Galí, López-Salido, and Vallés, 2007).

The empirical literature has devoted considerable attention to the identification of fiscal shocks. Barro (1981) and Ramey and Shapiro (1998) use approaches that exploit variation in government spending not directly driven by contemporaneous output. Blanchard and Perotti (2002) identify structural fiscal shocks using institutional information on tax and spending elasticities, and Perotti (2005) extends this framework across OECD economies. Perotti also documents substantial instability across subsamples, an important motivation for models with time-varying parameters.

Mountford and Uhlig (2009) propose an alternative identification strategy based on sign restrictions. Rather than relying on a recursive ordering or a single set of contemporaneous elasticities, sign restrictions discipline selected responses while leaving others unrestricted. This approach is particularly useful in the present application because the response of GDP to a government spending shock can be left unrestricted and therefore is not imposed by construction.

A separate literature emphasizes state dependence. Auerbach and Gorodnichenko (2012) use regime-switching methods and find that government spending is considerably more effective in recessions than in expansions. This result is directly relevant for the present paper: a constant-coefficient VAR may obscure economically meaningful variation in fiscal transmission over the business cycle. The published corrigendum to their paper reports that corrected data produce results very similar to those originally reported.

Time-varying parameter models provide another way to study this instability without forcing the economy into a fixed number of regimes. Kirchner, Cimadomo, and Hauptmeier (2010) estimate a TVP-VAR for the euro area and document substantial changes in the transmission of government spending shocks over time. Pereira and Lopes (2010) similarly emphasize the limitations of arbitrary subsample splits. Berg (2015) augments a TVP-VAR with fiscal forecasts and studies the determinants of the German fiscal multiplier, finding that fiscal sustainability and financial conditions matter for the magnitude of the multiplier. Glocker et al. (2017) also document time variation in fiscal transmission in the United Kingdom.

For Peru, Mendoza and Melgarejo (2008) estimate a structural VAR and find that fiscal policy was more powerful during 1990–2006 than during the 1980s. Sánchez and Galindo (2013) use a smooth-transition VAR and estimate larger spending multipliers in low-growth states than in high-growth states. Salinas and Chuquilín (2013) distinguish between current and capital spending and find stronger effects for capital expenditure. Together, these studies provide evidence that

fiscal transmission in Peru varies across states and periods, but they do not model its evolution as a continuous process. The TVP-VAR-SV framework used here is designed to fill that gap.

3 Empirical Methodology

The empirical analysis has three components. First, I estimate a measure of the Peruvian business cycle using the unobserved-components approach of Perron and Wada (2009), while reporting alternative filters as a sensitivity check. Rodríguez and Guillén (2014) show that this class of models provides a useful characterization of the Peruvian cycle. Second, I estimate the TVP-VAR-SV and identify structural shocks with sign restrictions. Third, I use the resulting impulse responses to construct time-varying fiscal multipliers and relate those multipliers to macroeconomic and fiscal conditions.

3.1 Fiscal multiplier definitions

Let $y_{h,t}$ denote the impulse response of real GDP at horizon h to a government spending shock identified at date t , and let $g_{h,t}$ denote the corresponding response of real government spending. Let Y_t and G_t denote the levels of real GDP and real government spending at date t . The estimation horizon is 20 quarters, and the time index covers the effective estimation sample from 1993Q3 to 2017Q2.

The impact multiplier is

$$M_{h,t}^I = \frac{y_{h,t}}{g_{0,t}} \frac{Y_t}{G_t}. \quad (1)$$

It converts the normalized impulse responses into units of output per unit of government spending.

The cumulative multiplier through horizon k is

$$M_{k,t}^C = \frac{\sum_{j=0}^k y_{j,t}}{\sum_{j=0}^k g_{j,t}} \frac{Y_t}{G_t}. \quad (2)$$

Finally, the present-value multiplier discounts future responses using the average interest rate i over the sample:

$$M_{k,t}^{PV} = \frac{\sum_{j=0}^k (1+i)^{-j} y_{j,t}}{\sum_{j=0}^k (1+i)^{-j} g_{j,t}} \frac{Y_t}{G_t}. \quad (3)$$

3.2 TVP-VAR-SV model

Following Cogley and Sargent (2005) and Primiceri (2005), consider the reduced-form VAR

$$\mathbf{z}_t = B_{0,t} + \sum_{\ell=1}^L B_{\ell,t} \mathbf{z}_{t-\ell} + u_t, \quad u_t \sim N(0, \Sigma_t), \quad (4)$$

where $\mathbf{z}_t$ is the vector of endogenous variables described in Section 5, $B_{0,t}$ is a vector of time-varying intercepts, $B_{\ell,t}$ contains time-varying lag coefficients, and Σ_t is the time-varying covariance matrix

of reduced-form innovations.

Following Primiceri (2005), the time-varying slope coefficients, contemporaneous coefficients, and log volatilities are collected in β_t , α_t , and $\ln(\sigma_t)$, respectively, and evolve as random walks:

$$\beta_t = \beta_{t-1} + \xi_t, \quad \xi_t \sim N(0, \Sigma_\beta), \quad (5)$$

$$\alpha_t = \alpha_{t-1} + \zeta_t, \quad \zeta_t \sim N(0, \Sigma_\alpha), \quad (6)$$

$$\ln(\sigma_t) = \ln(\sigma_{t-1}) + \eta_t, \quad \eta_t \sim N(0, \Sigma_\sigma). \quad (7)$$

This specification allows gradual changes in both propagation dynamics and shock volatility.

The model is estimated with Bayesian methods. The prior distributions are

$$\begin{aligned} \beta_0 &\sim N(\hat{\beta}, \hat{V}_\beta), \\ \alpha_0 &\sim N(\hat{\alpha}, \hat{V}_\alpha), \\ \ln(\sigma_0) &\sim N(\ln(\hat{\sigma}_0), I_n), \\ \Sigma_\beta &\sim W(s_1 k_1 \hat{V}_\beta, s_1), \\ \Sigma_\alpha &\sim W(s_2 k_2 \hat{V}_\alpha, s_2), \\ \Sigma_\sigma &\sim W(s_3 k_3 I_n, s_3). \end{aligned}$$

The prior means and covariance matrices are calibrated from an OLS training sample. The hyperparameters k_1 , k_2 , and k_3 govern the prior degree of time variation. The baseline uses $k_1 = 0.0005$, $k_2 = 0.05$, and $k_3 = 0.05$.

The baseline specification uses one lag. A constant-parameter VAR estimated with the same variables points to one or two lags, and the two-lag case is examined in the sensitivity analysis. Posterior simulation uses 10,000 Gibbs iterations, with the first 5,000 discarded as burn-in. Every tenth remaining draw is retained to reduce dependence among stored draws.

4 Identification Strategy

Structural shocks are identified using sign restrictions. The goal is to distinguish aggregate demand, aggregate supply, monetary policy, tax-revenue, government-spending, and external shocks while imposing as little structure as possible on the fiscal responses of primary interest. Table 1 reports the impact restrictions.

Table 1: Sign restrictions

	GDP	Tax Rev.	Gov. Spend.	Inflation	Int. Rate	S&P GSCI
Demand Shock	+			+		
Supply Shock	+			-		
Tax Shock		+				
Gov. Spending Shock			+			
Monetary Policy Shock	+			+	-	
Global Shock						+

Note: Signs are imposed only on impact. A “+” (“-”) indicates a positive (negative) impact response of the column variable to the row shock. Blank entries are unrestricted.

The restrictions are imposed only on impact; subsequent responses are determined by the estimated dynamics of the model. For aggregate supply and monetary policy shocks, the restrictions follow standard macroeconomic sign predictions. The external shock is identified through the S&P GSCI, which captures broad movements in global commodity prices and external conditions relevant for a small commodity-exporting economy such as Peru.

The fiscal identification is deliberately agnostic. A positive government-spending shock is required only to raise government spending on impact; the responses of GDP, tax revenue, inflation, and the interest rate are left unrestricted. Similarly, a positive tax-revenue shock is required only to raise tax revenue on impact. This design follows the spirit of Mountford and Uhlig (2009) and avoids imposing the sign of the output response to fiscal shocks by construction. Impulse responses are computed over a 20-quarter horizon at each date in the sample.

5 Data

The model uses six quarterly variables: real GDP, nonfinancial central-government spending, central-government tax revenue, the consumer price index (CPI), the S&P GSCI, and a short-term interest rate. The raw sample runs from 1992Q2 to 2017Q2. Real GDP is measured in millions of 2007 soles. Government spending and tax revenue are deflated by the CPI (2009=100), and real GDP, spending, and tax revenue are seasonally adjusted using Census X-13. Appendix 1 plots the series in levels and transformations.

The interbank interest rate is available only from 1995Q4 onward. To extend the series back to 1992Q2, I splice it with the BCRP certificate-of-deposit rate, which served as an operational instrument for liquidity management during the earlier period. This procedure may introduce measurement error around the splice; the Bayesian estimation does not, by itself, eliminate that source of uncertainty.

Table 2: Unit root tests: series in levels

	DF ^{GLS}	PP	Ng Perron	ERS	Zivot-Andrews
GDP	-1.16	-1.43	-1.54	18.9	-3.93
Tax Revenue	-1.37	-1.38	-1.34	18.6	-3.31
Expenditure	-2.08	-4.25	-1.99	11.8	-3.50
CPI	-1.68	-16.22	-0.16	1381	-5.04
SPGSCI	-1.85	-1.94	-1.78	12.9	-3.75
Interest Rate	-0.82	-5.37	-0.79	79.9	-6.86

Note: All tests include a constant and a deterministic trend. The Zivot–Andrews specification allows for a break in both the intercept and trend. Critical values are reported in the original estimation files; the table reports the corresponding test statistics.

Table 2 reports unit-root tests for the variables in levels. Taken together, the tests do not provide sufficiently strong evidence of stationarity for most level series, so the macroeconomic quantities enter the model in transformed form. The short-term interest rate is kept in levels because the PP and Zivot–Andrews tests reject the unit-root null and because the policy rate is economically expected to be mean reverting.

Table 3: Unit root tests: transformed series

	DF ^{GLS}	PP	Ng Perron	ERS	Zivot-Andrews
$\Delta_4 \log(\text{GDP})$	-2.53	-3.75	-2.06	2.91	-3.51
$\Delta_4 \log(\text{Tax Revenue})$	-3.17	-3.23	-2.87	1.59	-3.58
$\Delta_4 \log(\text{Expenditure})$	-3.15	-7.18	-3.18	0.77	-4.46
$\Delta_4 \log(\text{CPI})$	-1.13	-7.06	-0.59	208	-5.45
$\Delta_4 \log(\text{SPGSCI})$	-4.93	-3.68	-4.69	0.56	-4.17

Note: Δ_4 denotes the four-quarter difference. All tests include a constant; the CPI specification also includes a deterministic trend. The Zivot–Andrews specification allows for a break in the intercept. Critical values are reported in the original estimation files; the table reports the corresponding test statistics. The short-term interest rate enters the TVP-VAR-SV in levels and is therefore not reported in this table.

Table 3 reports the corresponding tests for the transformed series. The results support treating these variables as stationary. The transformed macroeconomic variables, together with the short-term interest rate in levels, therefore constitute the endogenous vector used in the TVP-VAR-SV.

The Peruvian economy underwent several large shocks and policy changes over the sample, making multiple structural breaks plausible. Table 4 applies the Bai–Perron methodology to the transformed series. The tests indicate multiple breaks in GDP growth, inflation, SPGSCI growth, and the interest rate, with break dates differing across variables. This heterogeneity reinforces the motivation for a model in which coefficients and volatility can evolve continuously rather than changing only at a small set of common break dates.

Table 4: Multiple structural-break tests: transformed series

	Δ GDP	Δ Tax Revenue	Δ Expenditure	Δ Inflation	Δ SPGSCI	Interest Rate
Sup $F_P(1-0)$	8.7*	35.7*	1.9	28.9*	9.7*	93.2*
Sup $F_P(2-1)$	22.2*			27.6*	11.1*	10.4*
Sup $F_P(3-2)$	36.9*			17.4*	4.8	7.8
Sup $F_P(4-3)$	9.4					
UD max (5%)	3	1	0	1	5	2
WD max (5%)	5	5	0	3	5	2
BIC	3	4	0	1	2	2
LWZ	2	0	0	1	0	2
$\hat{T}_1$	1998-Q1	2014-Q1		1996-Q4	1999-Q3	1996-Q1
$\hat{T}_2$	2001-Q4			2000-Q2	2009-Q1	2001-Q3
$\hat{T}_3$	2014-Q1			2007-Q4		

Note: The Bai–Perron tests are applied to annual rates with an intercept-only specification, a trimming parameter of 0.15, and a maximum of five breaks. An asterisk indicates that the corresponding Sup-F statistic exceeds the 5% critical value. The test for log SPGSCI in levels is omitted because it does not display a significant break.

6 Results

6.1 Evidence of Time Variation

Following Bijsterbosch and Falagiarda (2015), I use three diagnostics to assess whether the parameters of the model are stable over time. The Cogley–Sargent trace statistic compares the estimated innovation variance of the time-varying parameters with its posterior distribution. I also compare parameter distributions across selected dates using Kolmogorov–Smirnov and t -tests. The comparisons are 1994Q1 versus 2005Q1 and 2005Q1 versus 2015Q1.

The trace statistic is 0.05, below the reported posterior percentiles. The distributional tests also reject stability for a large fraction of the elements of the contemporaneous-coefficient matrix A , volatility block H , and lag-coefficient block B . Taken together, these diagnostics provide evidence of economically meaningful time variation and support the use of the TVP-VAR-SV specification.

Table 5: Changing Coefficient and Volatility Parameter Tests

Trace test		Kolmogorov–Smirnov test			t -test	
Trace	0.05		1994Q1–2005Q1	2005Q1–2015Q1	1994Q1–2005Q1	2005Q1–2015Q1
16%	0.91	A	42/42	28/42	41/42	29/42
50%	1.51	H	5/6	6/6	6/6	6/6
84%	2.72	B	14/15	11/15	13/15	15/15

Note: The Kolmogorov–Smirnov and t -tests compare parameter distributions across the indicated dates. Tests are applied separately to each element of A , H , and B . Numerators report the number of parameters for which equality is rejected at the significance level used in the original implementation (0.01%); denominators report the total number of parameters in each block.

6.2 Stochastic Volatility and Fiscal Impulse Responses

Appendix 3.A reports the estimated stochastic volatility of the model innovations. GDP volatility rises sharply around the late-1990s external and domestic disturbances and peaks in 1999, before declining during the 2000s. The increase around the global financial crisis is noticeably smaller than the late-1990s spike. This pattern is consistent with the greater macroeconomic stability documented for Peru during the 2000s (Yamada and Castro, 2010).

Tax-revenue and government-spending volatility display a less monotonic pattern. Revenue volatility increases during the commodity-price boom, while spending volatility rises sharply around the global financial crisis. By contrast, inflation volatility falls markedly over the sample, consistent with the stabilization that followed the high-inflation period and the adoption of inflation targeting. The volatility of the short-term interest rate also declines after the monetary-policy regime shifts toward explicit interest-rate management. The S&P GSCI exhibits the largest global-crisis spikes among the variables in the system.

Because the model is time varying, an impulse response is estimated for every date in the effective sample. To make responses comparable across dates, I normalize the government-spending innovation so that spending rises by 1% on impact in each period.

Under the agnostic fiscal identification, a positive government-spending shock generates a positive output response on average (Figure 5, Appendix 3.B). A 1% increase in public spending raises GDP by about 0.18% on impact, and the 68% posterior credible interval excludes zero for roughly two quarters. The response of tax revenue is less precisely estimated. One possible interpretation is that Peru's high degree of informality weakens the short-run mapping from additional economic activity into tax collection, although the model does not identify that mechanism directly.

Figure 6 reports the response to a positive tax-revenue shock. GDP rises on impact, which differs from the textbook contractionary response to an exogenous tax increase. Because the identification leaves GDP unrestricted, the result should be interpreted as a feature of the identified revenue innovation rather than as evidence that an exogenous tax increase raises output. Revenue innovations may combine changes in the tax base, enforcement, and policy packages correlated with other fiscal actions. The present model does not separately identify these channels, so I do not assign a causal interpretation to this response.

Appendix 3.C shows how the impact response to a normalized spending shock evolves over time. The estimated GDP response increases through the second half of the 1990s, reaching roughly 0.44% near the end of 1998, and then declines after 1999 toward values close to 0.1% in the later part of the sample. The timing of the decline coincides with the consolidation of major trade, financial, and fiscal reforms, but the TVP-VAR does not identify those reforms as the cause of the change. I therefore treat this timing as descriptive rather than causal.

This time variation helps reconcile the estimates with Mendoza and Melgarejo (2008). Over a comparable 1990–2006 window, the average impact response from the present model is close to the magnitude they report. The TVP-VAR-SV, however, reveals that the average masks substantial within-period variation: relatively large responses in the late 1990s are followed by a persistent decline after 1999. This pattern is also consistent with international evidence that fiscal transmission

can change substantially over time, including Kirchner et al. (2010).

6.3 Spending Multipliers and Their Determinants

Appendix 4 reports the time-varying spending multipliers. The impact multiplier closely tracks the normalized GDP response. It reaches its highest values during 1997–2000, when the estimate is around 2: one sol of additional government spending is associated with roughly two soles of additional output on impact. The impact multiplier then declines, falling below one after the mid-2000s and averaging roughly 0.68 in the later part of the sample.

The cumulative and present-value multipliers show a less pronounced decline. At medium and long horizons, both measures remain above the corresponding short-run multiplier for much of the sample. During 2005–2017, the longer-horizon estimates fluctuate around 1.5. These results suggest that the post-1999 decline is concentrated more strongly in the immediate output response than in the accumulated response over several quarters.

I next examine which macroeconomic and fiscal conditions are associated with the estimated multipliers. Because the multiplier is itself estimated in the first stage, and because the regressors are not generated by exogenous variation, these regressions are descriptive. They are useful for evaluating whether the comovement of the multiplier with observable conditions is consistent with mechanisms emphasized in the literature, but they should not be interpreted as causal estimates.

Auerbach and Gorodnichenko (2012) find that government spending is considerably more effective in recessions than in expansions. I therefore include the business-cycle component estimated from the Perron and Wada (2009) model. I also include the spread between lending and deposit rates as a measure of financial conditions. Berg (2015) finds that financial conditions are an important source of time variation in fiscal multipliers in Germany, although his estimates point to smaller multipliers during periods of financial-market stress; I therefore treat the sign of the spread coefficient as an empirical question in the Peruvian data.

Credit growth is included as a proxy for the tightness of private financing constraints. In models with liquidity-constrained households, easier access to credit can reduce the fraction of agents whose spending responds strongly to current income (Galí, López-Salido, and Vallés, 2007). Public debt-to-GDP is included to capture the possibility that fiscal space, policy credibility, or expectations of future adjustment affect fiscal transmission (Ilzetzki et al., 2013). Finally, trade openness and the savings rate capture two standard leakages from domestic demand: imports and private saving.

The second-stage analysis distinguishes the impact multiplier from the longer-horizon cumulative and present-value multipliers. All covariates enter with one lag, and the regressions include a constant and a time trend. Lagging the regressors reduces contemporaneous feedback but does not solve endogeneity or reverse-causality concerns. Newey–West corrections are used for heteroskedasticity and serial correlation, and coefficients are standardized to facilitate comparison across covariates.

Table 6: Determinants of fiscal multipliers

	Impact Multiplier	Cumulative Multiplier	Present Value Multiplier
Business Cycle	-0.08	-0.51***	-0.51***
Interest Rate Spread	0.20	0.11	0.15
Credit Growth	-0.30**	-0.44**	-0.41*
Debt/GDP Ratio	-1.61**	-1.56**	-1.90***
Trade Openness	-0.94***	-0.24	-0.43*
Savings Rate	-0.07***	0.06	0.02**
Constant	8.80***	5.54***	6.17***
Trend	-0.02***	-0.02***	-0.02***
Adjusted R-squared	0.73	0.59	0.63

Note: The regressions include a constant and a linear time trend, and all covariates enter with one lag. Newey–West corrections account for heteroskedasticity and serial correlation, with the lag length selected by the AIC. Reported coefficients are standardized to facilitate comparisons across determinants. ***, **, and * denote statistical significance at the 1, 5, and 10 percent levels, respectively.

Table 6 reports the second-stage estimates. The business-cycle coefficient is negative in all three specifications. Given the construction of the cycle variable, the sign is consistent with a smaller multiplier in expansions and a larger multiplier in downturns. The association is modest at impact and larger in absolute value at longer horizons. Alternative measures of the business cycle reported in Appendix 2 generate the same qualitative pattern. This evidence is consistent with the state-dependent multiplier literature, including Auerbach and Gorodnichenko (2012), but the present regressions are not designed to identify a causal state effect.

The coefficient on the lending–deposit spread is positive but imprecisely estimated. Thus, unlike Berg’s (2015) German results, the Peruvian estimates do not provide clear evidence that tighter financial conditions reduce the multiplier.

Credit growth is negatively associated with the estimated multiplier. One interpretation is that stronger private credit conditions relax household and firm financing constraints, reducing the sensitivity of private demand to fiscal stimulus. This mechanism is consistent with models featuring liquidity-constrained agents, but the coefficient should be interpreted as suggestive rather than causal.

The debt-to-GDP ratio has a negative coefficient in all three specifications and is one of the largest associations in standardized terms. This pattern is consistent with the idea that fiscal space and policy credibility may matter for fiscal transmission. The estimates do not establish that reducing public debt would mechanically raise the fiscal multiplier: debt is endogenous and may proxy for other macroeconomic or institutional conditions. I therefore interpret the result as an empirical association that motivates further causal analysis rather than as a direct policy prescription.

Trade openness is negatively associated with the impact multiplier, consistent with leakage of additional demand into imports. The savings-rate coefficient is also negative in the impact specification. These relationships are less stable at longer horizons, reinforcing the need to interpret the

second-stage estimates as descriptive correlations.

6.4 Sensitivity Analysis

I assess sensitivity along three dimensions. First, I re-estimate the model under alternative prior specifications. One alternative uses the Primiceri (2005) calibration, $k_1 = 0.01$, $k_2 = 0.01$, and $k_3 = 0.1$. A second uses the calibration in Castillo, Montoya, and Quineche (2016), $k_1 = 0.01$, $k_2 = 1$, and $k_3 = 0.1$. Figure 10 in Appendix 5 shows that the main volatility, impulse-response, and multiplier patterns are similar across these prior choices.

Second, I increase the lag length from one to two. Third, I replace the S&P GSCI with the terms of trade. Figure 11 reports these exercises. The alternative specifications generate some quantitative differences, particularly near the beginning of the sample, but the main qualitative patterns in stochastic volatility, the average GDP response, the time-varying impact response, and the spending multiplier are preserved.

7 Conclusions

This paper documents substantial time variation in the effects of public spending in Peru. Using a TVP-VAR-SV identified with sign restrictions, I find that the short-run GDP response to a normalized government-spending shock increased through the late 1990s and declined after 1999. The resulting impact multiplier follows the same pattern, while cumulative and present-value multipliers display a less pronounced decline at longer horizons. Stability diagnostics for the model parameters support the use of a time-varying specification rather than a constant-coefficient VAR.

The second-stage analysis asks which observable macroeconomic and fiscal conditions covary with the estimated multipliers. Higher public debt-to-GDP ratios are associated with smaller multipliers, and the business-cycle estimates are consistent with larger multipliers in weaker states of the economy. Credit growth and trade openness are also negatively associated with some multiplier measures. These relationships are informative about possible mechanisms, but they are not causal estimates: the dependent variables are generated in the first stage, the regressors are endogenous, and lagging them does not eliminate reverse causality.

The main contribution of the paper is therefore twofold. First, it provides a continuous estimate of how fiscal transmission in Peru changed over more than two decades, rather than imposing fixed subsamples or a small number of regimes. Second, it links that time variation to observable macroeconomic conditions while making explicit the limits of the second-stage identification. A natural extension is to develop a framework that jointly accounts for first-stage estimation uncertainty and introduces exogenous variation in the proposed determinants of the fiscal multiplier.

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Appendix 1

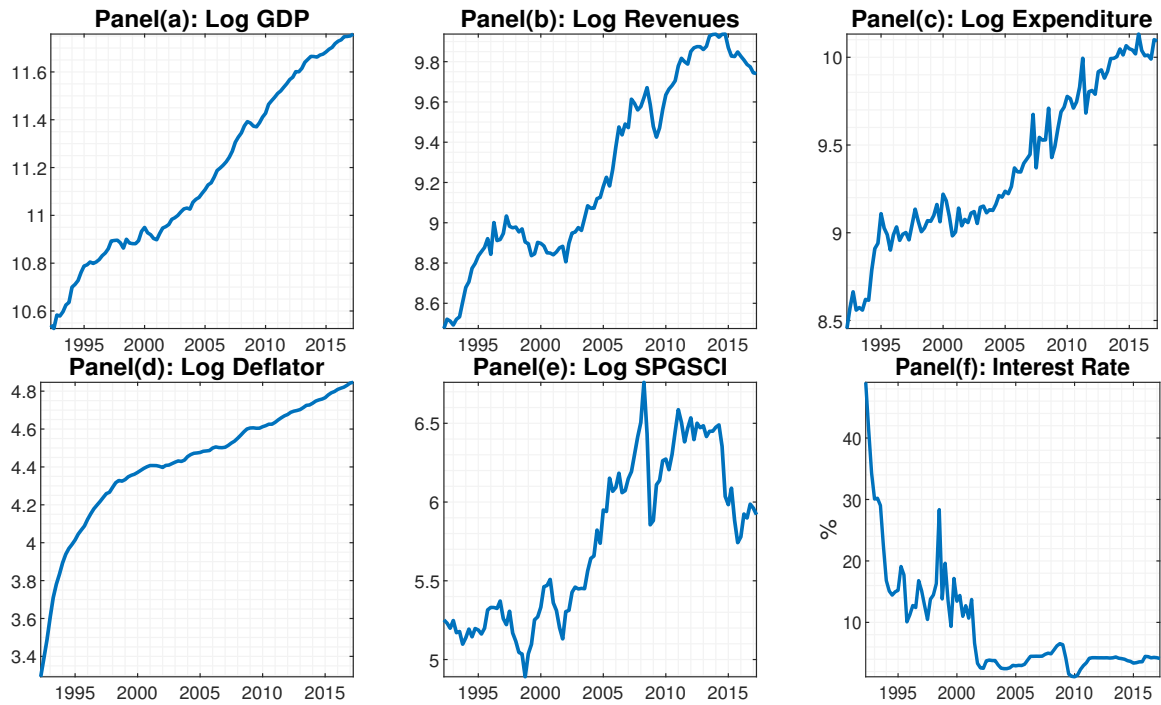


Figure 1: Series in levels. Real GDP, government spending, and tax revenue are seasonally adjusted using Census X-13. The short-term interest rate is reported in levels. Source: BCRP.

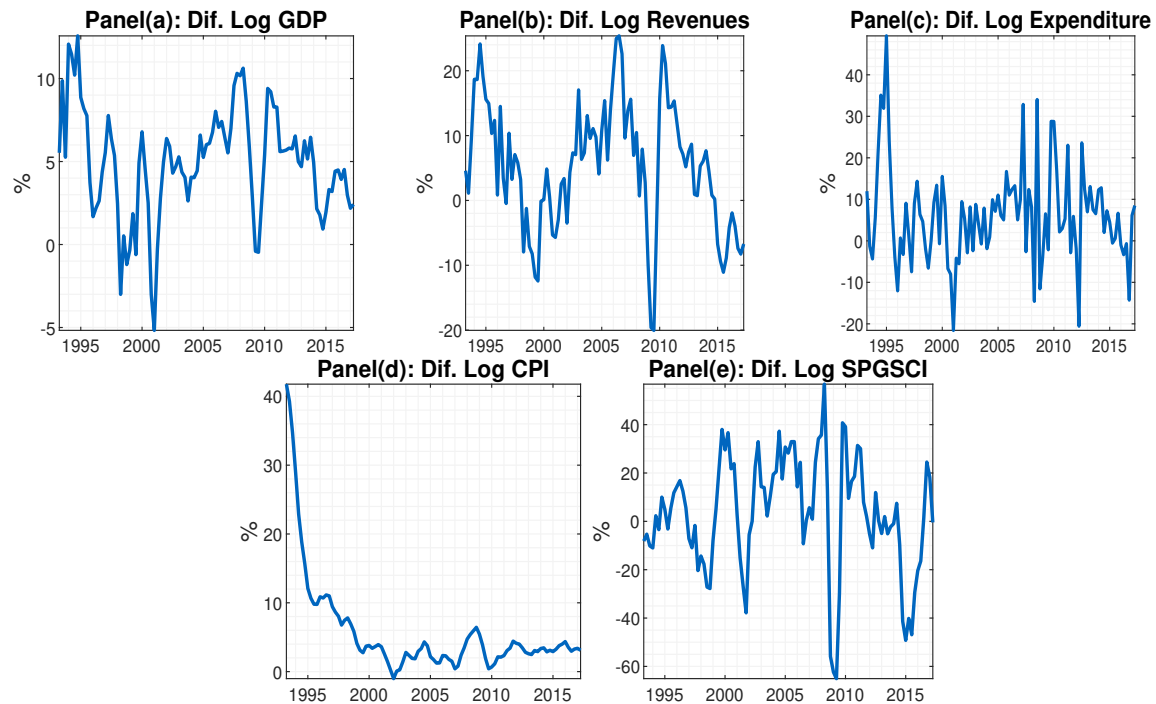


Figure 2: Transformed series. Annual differences are computed relative to the same quarter of the previous year. The short-term interest rate remains in levels. Source: BCRP.

Appendix 2

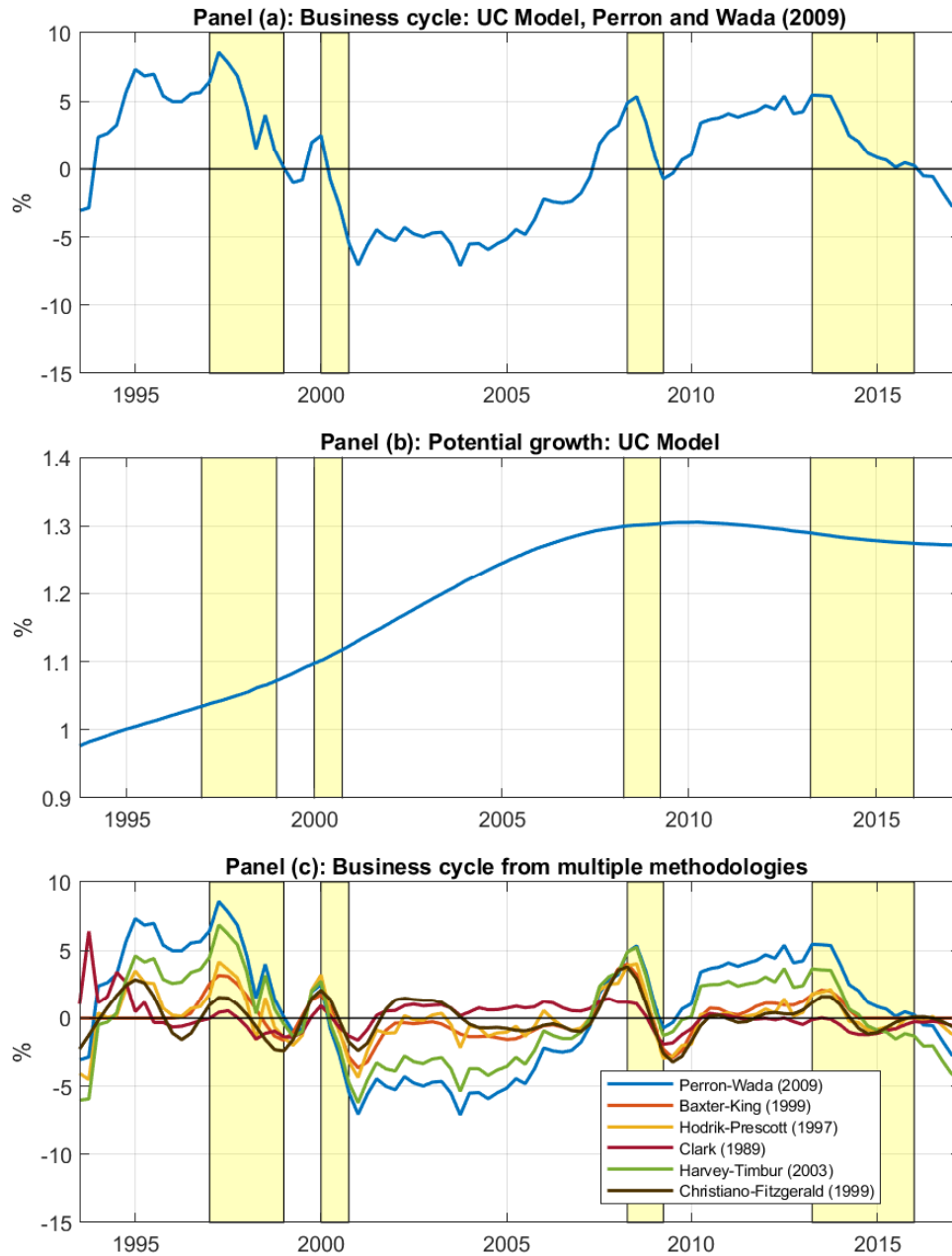


Figure 3: Business-cycle estimates. Panel (a) reports the cycle from the unobserved-components (UC) model with an endogenous break and mixture-normal innovations. Panel (b) reports estimated trend growth. Panel (c) compares the UC estimate with alternative filters commonly used in macroeconomics. Shaded areas denote recession periods identified by the UC model.

Appendix 3.A

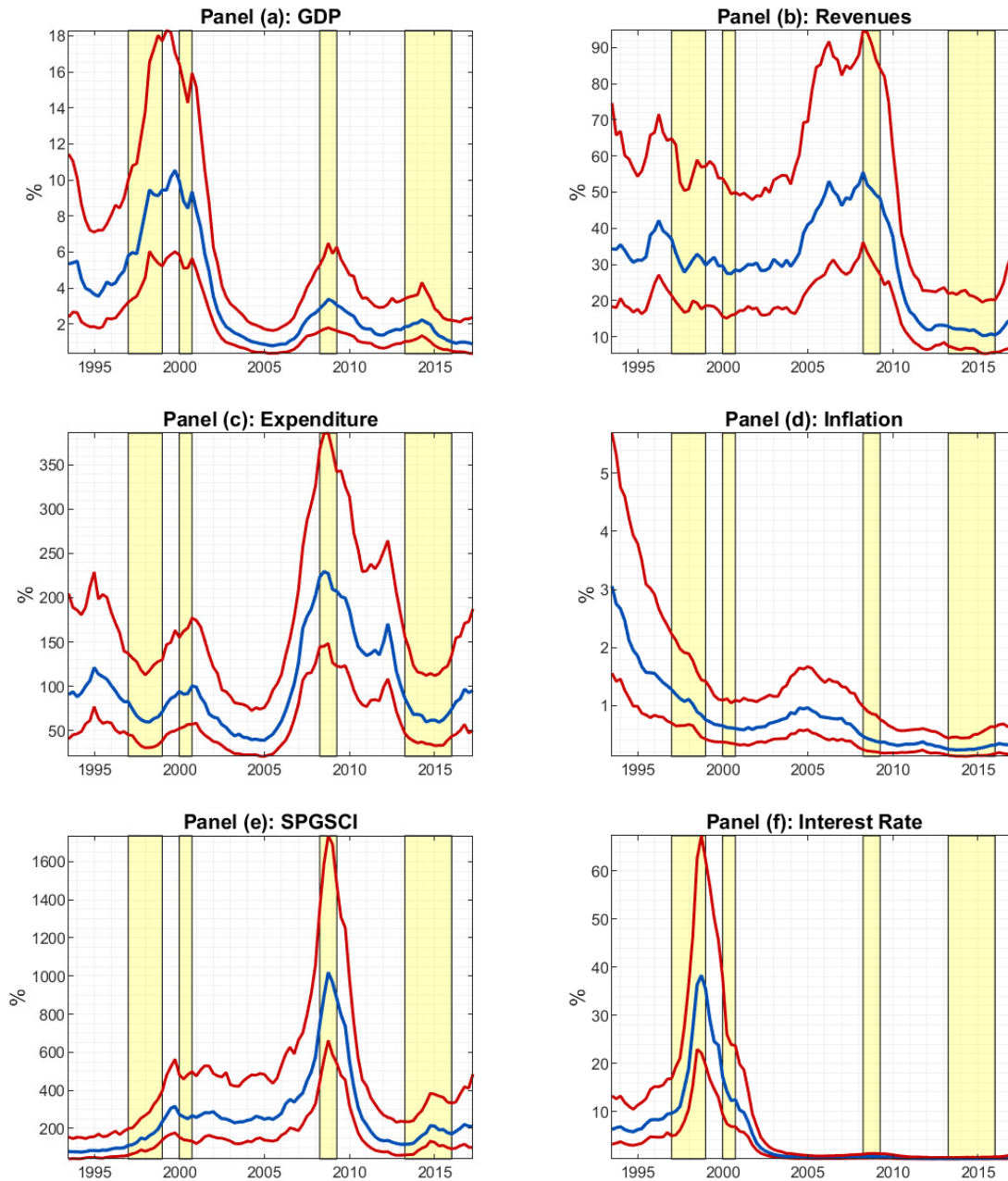


Figure 4: Stochastic volatility. Red lines denote the 16th and 84th percentiles of the posterior distribution and the blue line denotes the posterior median. Shaded areas identify recession periods from the UC model.

Appendix 3.B

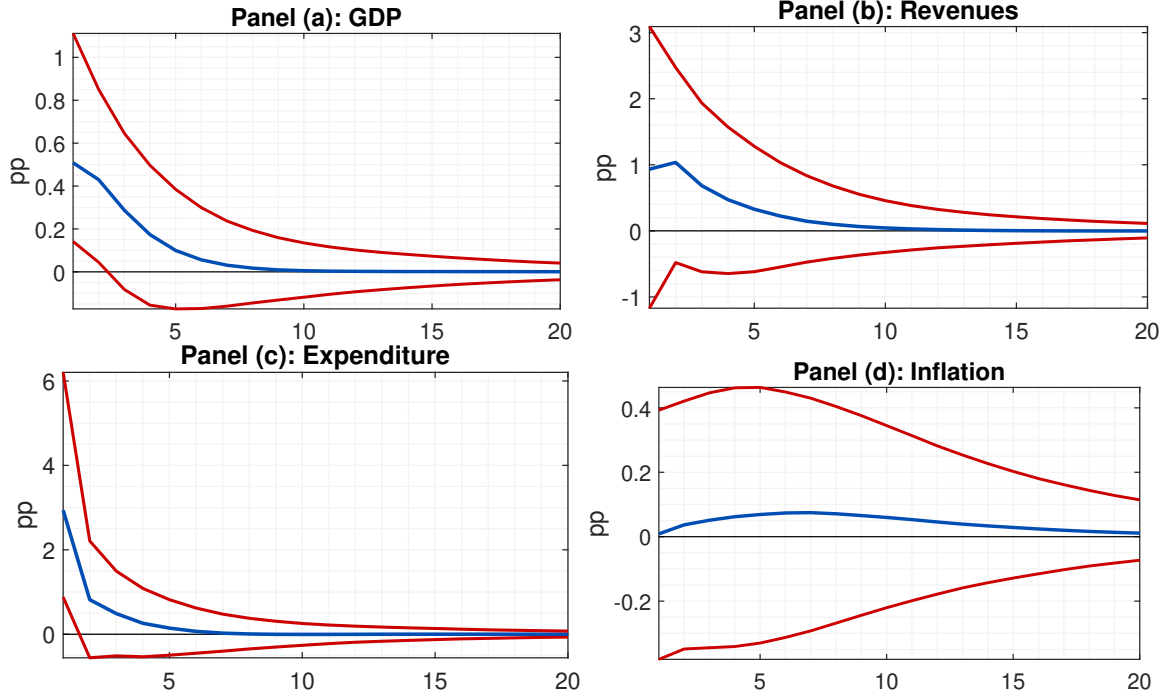


Figure 5: Government-spending shock. Red lines denote the 16th and 84th posterior percentiles and the blue line denotes the posterior median. The impulse-response horizon is 20 quarters.

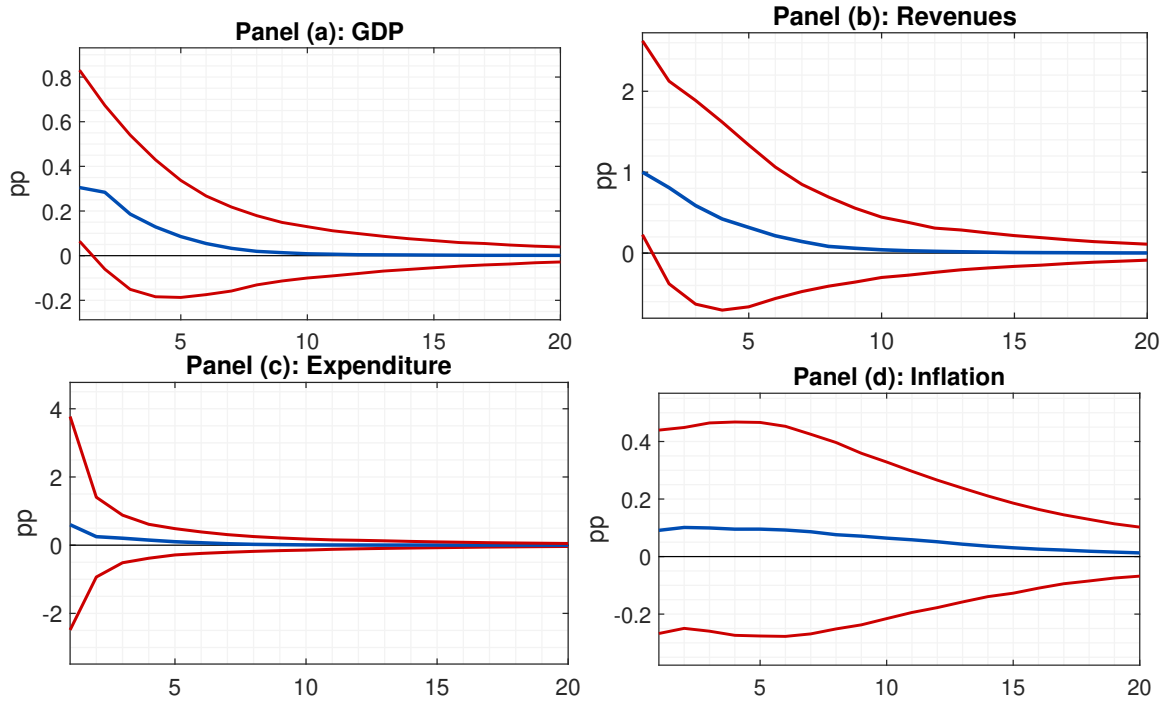


Figure 6: Tax-revenue shock. Red lines denote the 16th and 84th posterior percentiles and the blue line denotes the posterior median. The impulse-response horizon is 20 quarters.

Appendix 3.C

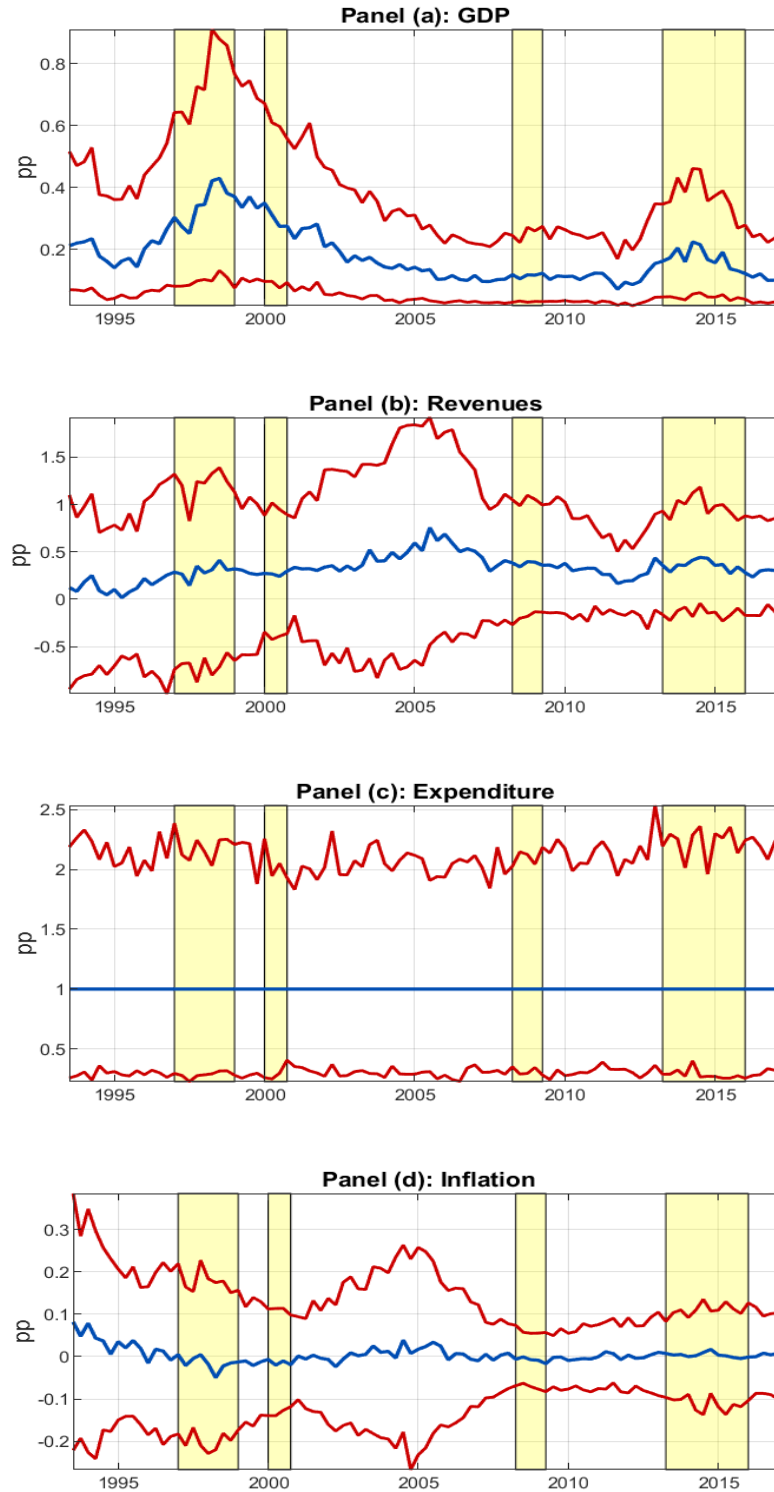


Figure 7: Time-varying impact responses to a normalized government-spending shock. Red lines denote the 16th and 84th posterior percentiles and the blue line denotes the posterior median. The spending shock is normalized to 1% in every period. Shaded areas identify recession periods from the UC model.

Appendix 4

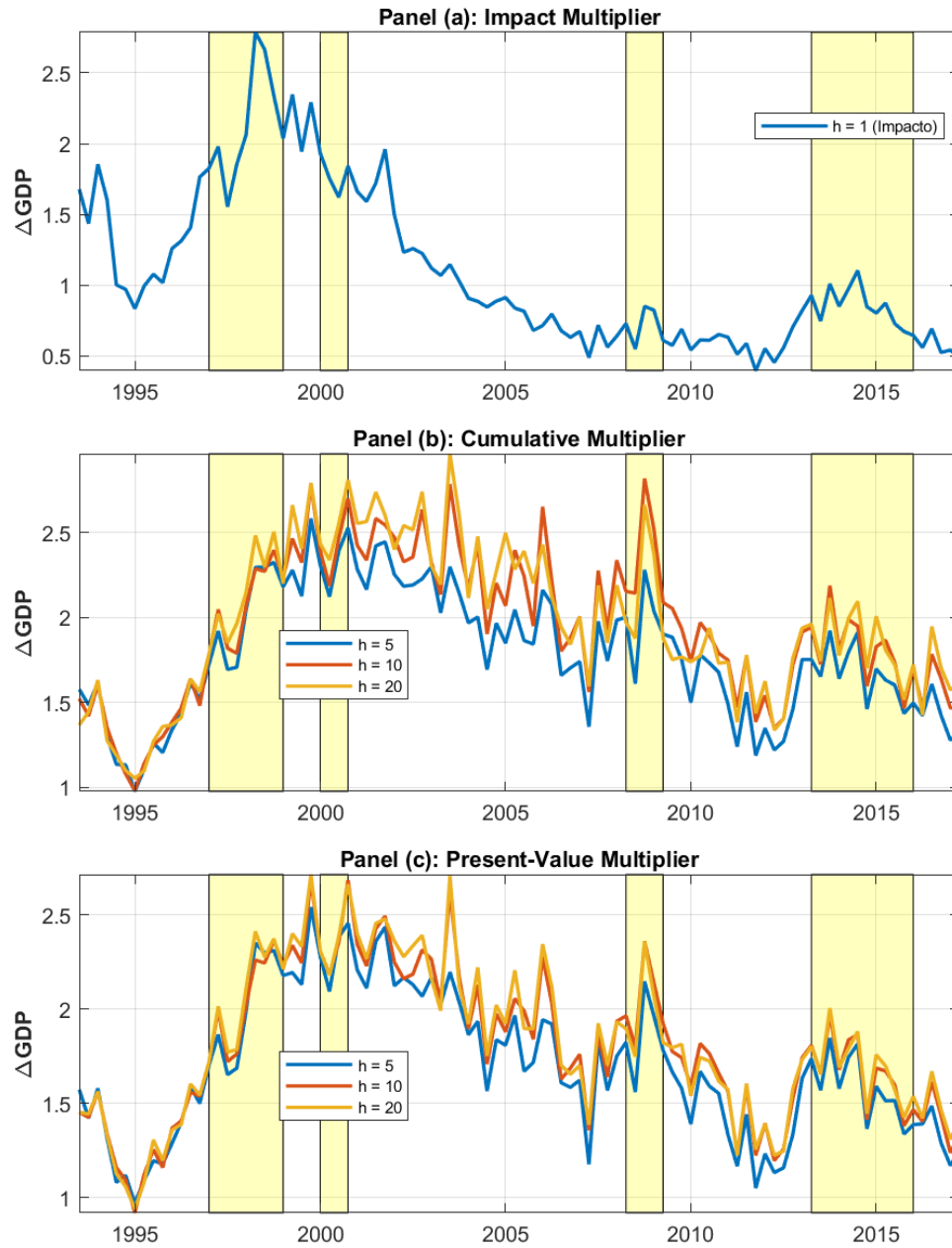


Figure 8: Spending multipliers. Panel (a) reports the impact multiplier. Panels (b) and (c) report the cumulative and present-value multipliers at horizons $h = 5$, $h = 10$, and $h = 20$. Lines show posterior medians. Shaded areas identify recession periods from the UC model.

Appendix 5

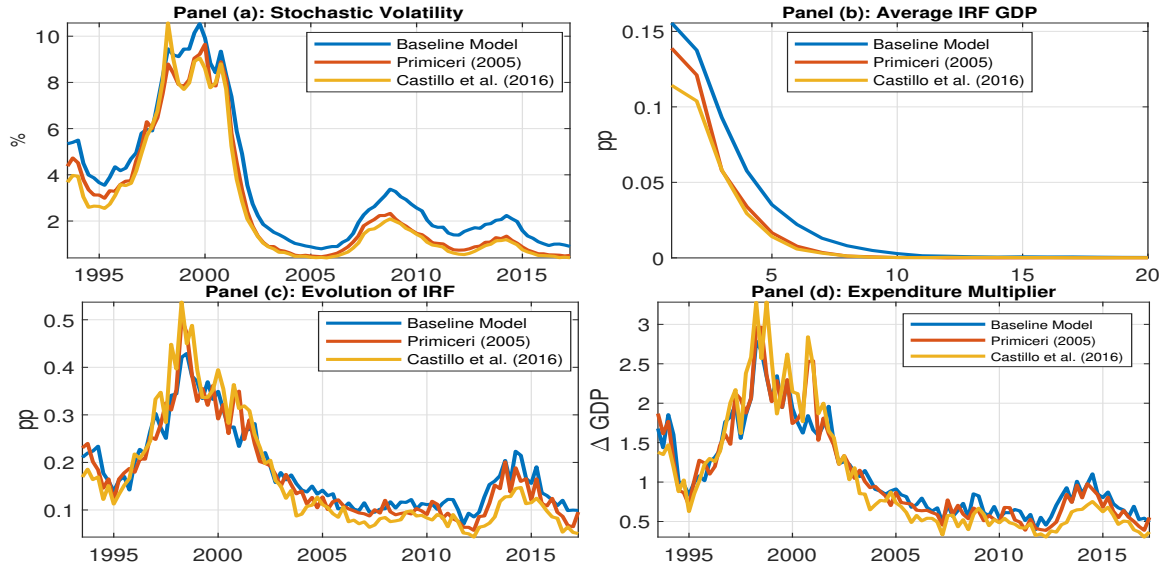


Figure 10: Sensitivity to prior specifications. The panels compare the baseline with the prior calibrations of Primiceri (2005) and Castillo et al. (2016), reporting GDP stochastic volatility, the average impulse response, the time-varying normalized impact response, and the spending multiplier.

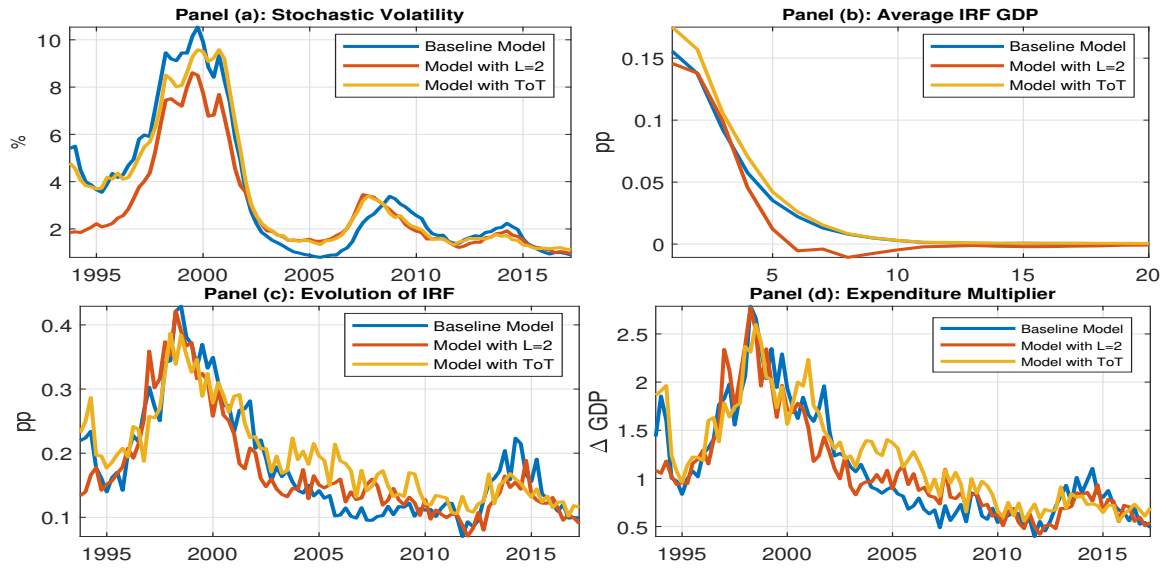


Figure 11: Sensitivity to lag length and the external variable. The panels compare the baseline with a two-lag specification and a specification that replaces the S&P GSCI with the terms of trade, reporting GDP stochastic volatility, the average impulse response, the time-varying normalized impact response, and the spending multiplier.